MOHAMMAD RAOUF ABEDINI

AI Security Researcher · Adversarial Robustness & AI Control · Empirical Safety Evaluation

Sydney · Relocating to San Francisco, CA · Visa sponsorship required
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PROFESSIONAL SUMMARY

Security researcher focused on keeping tool-using AI agents safe and beneficial: builds host-side controls for what an agent is permitted to do and stress-tests them with matched-control experiments against live models. Sole author of Project Simurgh, an open-source agent-authorization prototype whose October 2026 study reports a benefit on one model and a null result on another. Evaluated Claude for exploitable code and guardrail circumvention in Anthropic's safety-evaluation program via Alignerr. Five sole-author preprints and two coordinated vendor disclosures (Microsoft, Apple) after seven years in security operations and engineering. Available full-time from December 2026 for a three- to six-month San Francisco placement.

RESEARCH & TECHNICAL SKILLS

Empirical AI safety and alignment: matched-control experiment design · adversarial robustness and AI control for tool-using agents · prompt-injection and authority containment · guardrail and classifier evaluation · uncertainty reporting · host-held human-oversight state (delegation, stop, lease)
Large language models (LLMs) and agent tooling: Claude API and Claude Code · OpenAI Responses API with tool calling · self-hosted Llama (3.2-3B, 3.3-70B) · Llama Guard 4 · AgentDojo · MCP servers
AI security, safeguards and red-teaming: guardrail-circumvention analysis · red-team scenario design · vulnerability research and coordinated disclosure · offensive proofs of concept · intrusion detection · phishing-detection ML · incident response · MITRE ATT&CK
Languages and systems: Python · TypeScript · C · C++17/20 · Rust · Kotlin · Lean 4 · SQL · Bash · Docker · GitHub Actions · PostgreSQL · Linux · eBPF

RESEARCH & OPEN SOURCE

Project Simurgh — Inspectable Authorization and Containment for AI Agents (Sole Author) May 2026 – Present
github.com/Raoof128/Project-Simurgh · 92 releases · software DOI 10.5281/zenodo.21737792

- **Designed and ran a transfer study (Oct 2026):** 24 tasks, two trials each, three matched controls (credential-only, composed, deny-all) and two models; 288 replayed branches. On Meta Llama-3.2-3B, composed host controls cut restricted-state trials with new effects from 17/24 to 0/24, kept 23/24 useful completions and blocked no valid proposal. On OpenAI gpt-6-luna the baseline was already 0/24, so the study reports no incremental protection there. Protocol to published package took three days (15–17 Oct 2026).
- **Reported uncertainty per cluster:** the 24 restricted trials fall into six states, so intervals are per state (Wilson 95%: 43.6–97.0% versus 0.0–39.0%); part of the gap holds by construction.
- **Detected and mitigated a defect in the live OpenAI integration:** an eight-case Responses API workflow exposed a role-placement error (8/8 declined). Retained the original traces; a frozen developer-role correction completed 2/3 authorized workflows and declined 5/5 restricted ones, with one refusal still open.
- **Captured a live Llama Guard 4 12B input classifier** over a 180-case set (frozen stage, 2026) and produced containment evidence for all 138 malicious cases it missed, 120 of them downstream injections outside an input-only classifier's view; a coverage claim, not a ranking of classifiers.
- **Reduced a live Llama-3.3-70B agent's targeted attack success** on AgentDojo from 9/140 to 0/140 over pre-registered injections in the workspace suite (frozen stage, 2026); the full four-suite deterministic run scored 0/949 attack success with 949/949 unattacked utility (both scoped as small experiments).
- **Built the work to be reproduced:** Ed25519-signed evidence archives replay offline in one command; 39 Lean 4 proofs sit behind a gate that fails on any unproven “sorry” placeholder; 5,171 tests passed in the 19 Oct 2026 smoke run.
- **Audited 919 of the project's own documents** and published all 141 dated findings, including overstatements in earlier closeouts, in the README.

The Invisible Window — Screen-Capture Evasion in WebRTC Proctoring (Sole Author) Feb 2026 – May 2026
github.com/Raoof128/invisible-window-research · IEEE-format preprint v2.0 · DOI 10.5281/zenodo.20376495

- Found that documented display-affinity APIs on Windows and macOS let a window stay visible on the monitor while contributing zero pixels to screen capture: 100% evasion of WebRTC proctoring on every platform tested, including macOS 26.
- Disclosed to Microsoft MSRC (Feb 2026) and Apple Product Security (Mar 2026) before publication; both classified the behavior as by design. Proof-of-concept code is withheld on dual-use grounds.

Aion-BibleQA — Retrieval and Citation Faithfulness in a RAG System (Sole Author) Jun 2026
Preprint · DOI 10.5281/zenodo.20522874

- Built and scored a 40-question benchmark for retrieval quality, citation support and false-premise robustness: R@5 of 0.941, mean citation support of 0.978, zero unsupported citations and 6/6 false-premise refusals.

Further Preprints and Shipped Security Tooling 2024 – 2026

- **Banking Shield** (DOI 10.5281/zenodo.20675513): machine-checked proofs that privacy-sensitive values never reach AI-generated explanations; five-tester dry run, zero leaks.
- **Simurgh device-integrity proofs** (DOI 10.5281/zenodo.20374849): preprint on privacy-preserving device integrity proofs for high-stakes sessions.
- **Student-society event pilot** (DOI 10.5281/zenodo.20549736): 31 consented sessions; ballot choices excluded from records; HMAC-chained audit logs.
- **Mehr Guard:** offline QR and URL phishing scanner in Kotlin Multiplatform; ensemble ML plus 25 heuristics at 87% F1 and under 5 ms, 19-scenario red-team mode.
- **Systems work (Rust, C++):** SentinelSSH, a Rust MCP server that gives AI agents policy-scoped, read-only inspection of Linux hosts through 17 typed tools and no shell; SentinelFlow, C++17 intrusion detection (500,000+ packets/sec, synthetic single-thread benchmark); Project Zurvan, local-first knowledge engine serving claims and decisions to agents over MCP (218 tests).

PROFESSIONAL EXPERIENCE

AI Safety Evaluator (Claude) — Alignerr Jan 2026 – Mar 2026
Anthropic safety-evaluation program · Remote

- Stress-tested Claude's safety guardrails: assessed outputs for exploitable code and harmful patterns, analyzed circumvention methods, and delivered rubric-based findings.

Cofounder — Syllabus-Sync May 2026 – Present
Macquarie University Incubator startup · Sydney · syllabus-sync.app

- Led the security backend (WebAuthn passkeys, TOTP MFA, Postgres row-level security, tamper-resistant audit logging) of a shared Next.js/TypeScript codebase that converts university syllabi into structured data; 503 tests across 92 files.

Security Engineer & Full-Stack Developer — Self-Employed (Freelance) Jan 2024 – Present
Sydney

- Delivered secure web applications for commercial clients with OWASP Top 10 controls and OAuth 2.0; 5.0-star rating across 31 Airtasker reviews.

IT Manager & Security Operations — Iran Pharmacy Aug 2019 – Jan 2024
50+ staff · Iran

- Investigated unauthorized-access incidents, built automated detection workflows that cut manual triage by about 30%, and enforced least-privilege access.

EDUCATION

Bachelor of Cyber Security — Macquarie University, Sydney Expected Jun 2027
Final session Feb–Jun 2027 is fully online · Coursework: Natural Language Processing & Machine Learning

Diploma of Information Technology — Macquarie University 2023 – 2024

CERTIFICATIONS & CREDENTIALS

Anthropic (Jul 2026), 20 courses including: Building with the Claude API · Claude Code in Action · Model Context Protocol

Macquarie University micro-credentials (2026, 93–100%): Security of AI · Application of AI · Data Security & Information Privacy · DevSecOps · Identity & Access Management